

Send command components**BaotouFD FC FB FA****Data length in frame (two bytes)**Intra-frame data (Command value two bytes +Parameter value*Nbyte) **End**

of package04 03 02 01

Receive command components**BaotouFD FC FB FA****Data length in frame (two bytes)****Return value (two bytes)** 00 indicates success, others indicate failureIntra-frame data (Return command value two bytes +Return value*Nbyte) **End**

of package04 03 02 01

Command Value:

Command Type	Command value (two bytes)	Command Data*N
Module opens command mode	0xFF	2Byte host computer version, default01
Module shutdown command mode	0xFE	none
Read the version number	0x00	none
Restart module	0x68	none
Read module configuration parameters	0x08	(2Byte parameter name) *NParameter names see table2
Set module configuration parameters	0x07	(2Byte parameter name +4Byte parameter value) * NParameter name and value see table2
Configuring system parameters	0x12	(2Byte parameter name +4Byte parameter value)Parameter name and value see table3

surface1

Parameter name:

Command Type	Parameter name (two bytes)	Parameter value (4byte)
Minimum detection distance door	0x00	scope0x00-0x0F
Maximum detection distance door	0x01	scope0x00-0x0F
Delay time	0x04	scope0x00-0xFF
Trigger threshold	0x10-0x1F	scope0-65536
Keep Threshold	0x20-0x2F	scope0-65536

surface2

System parameter name:

Command Type	Parameter name (two bytes)	Parameter value (4byte)
SystemMode	0x00	0x00 -Debug mode, in this mode the serial port reportsRDMaP(Parsed by the host computer) See the appendix for the parsing format4 0x04 -Reporting mode, in this mode, the serial port reports the energy value and detection results of each distance gate (to be parsed by the host computer). The parsing format is shown in the appendix surface5 0x64 -Running mode, in which the serial port prints out the status

surface3

Command instructions and parameter values are all small fields first

For example: Restart module command 68, command value 0x68 0x00

example:

Query trigger threshold 00 The value of , send: FD FC FB FA 04 00 08 00 10 00 04 03 02 01

return: FD FC FB FA 08 00 08 01 00 00 40 9C 00 00 04 03 02 01

Analysis:

Baotou FD FC FB FA Parameter data length 08 00 Return command value 08 01 Parameter Value 40 9C 00 00 End of package 04 03 02 01

The trigger threshold value is read as 16 Binary, small field first, 9C 40, so the value is 60000.

Sample instructions

Open command mode FD FC FB FA 04 00 FF 00 01 00 04 03 02 01

reply: FD FC FB FA 08 00 FF 01 00 00 02 00 20 00 04 03 02 01

Turn off command mode FD FC FB FA 02 00 FE 00 04 03 02 01

reply: FD FC FB FA 04 00 FE 01 00 00 04 03 02 01

Read the version number FD FC FB FA 02 00 00 00 04 03 02 01

reply: FD FC FB FA 0C 00 00 01 00 00 06 00 76 31 2E 35 2E 34 04 03 02 01

Restart module FD FC FB FA 02 00 68 00 04 03 02 01

Configure system mode (serial port status output mode) FD FC FB FA 08 00 12 00 00 00 64 00 00 00 04 03 02 01

reply: FD FC FB FA 04 00 12 01 00 00 04 03 02 01

Configure system mode (energy value output mode) FD FC FB FA 08 00 12 00 00 00 04 00 00 00 04 03

02 01 reply: FD FC FB FA 04 00 12 01 00 00 04 03 02 01

Configure system mode (debug output mode) FD FC FB FA 08 00 12 00 00 00 00 00 00 00 04 03 02

01 reply: FD FC FB FA 04 00 12 01 00 00 04 03 02 01

Read module parameters (minimum distance) FD FC FB FA 04 00 08 00 00 00 04 03 02 01

reply: FD FC FB FA 08 00 08 01 00 00 00 00 04 03 02 01

Set module parameters (minimum distance 00) FD FC FB FA 08 00 07 00 00 00 00 00 00 04 03 02 01

Successful reply: FD FC FB FA 04 00 07 01 00 00 04 03 02 01

Read module parameters (maximum distance) FD FC FB FA 04 00 08 00 01 00 04 03 02 01

reply: FD FC FB FA 08 00 08 01 00 00 0C 00 00 00 04 03 02 01

Set module parameters (maximum distance 12) FD FC FB FA 08 00 07 00 01 00 0C 00 00 00 04 03 02 01

Successful reply: FD FC FB FA 04 00 07 01 00 00 04 03 02 01

Read module parameters (delay time)FD FC FB FA04 0008 00 04 0004 03 02

01 reply:FD FC FB FA08 0008 01 0000 1E 00 00 0004 03 02 01

Set module parameters (delay time)FD FC FB FA08 0007 00 04 001A 00 00 0004 03 02 01

Successful reply:FD FC FB FA04 0007 01 0000 04 03 02 01

Read module parameters (hold threshold00) FD FC FB FA04 0008 00 20 0004 03 02

01 reply:FD FC FB FA08 0008 01 0000 40 9C 00 0004 03 02 01

Set the module parameters (hold threshold00) FD FC FB FA08 0007 00 20 0050 C3 00 0004 03 02 01

Successful reply:FD FC FB FA04 0007 01 0000 04 03 02 01

Read module parameters (trigger threshold00) FD FC FB FA04 0008 00 10 0004 03 02

01 reply:FD FC FB FA08 0008 01 0000 40 9C 00 0004 03 02 01

Set module parameters (trigger threshold00) FD FC FB FA08 0007 00 10 0050 9C 00 0004 03 02 01

Successful reply:FD FC FB FA04 0007 01 0000 04 03 02 01

Read module parameters (trigger threshold00-05)

FD FC FB FA0E 0008 00 10 00 11 00 12 00 13 00 14 00 15 0004 03 02 01

Successful reply:FD FC FB FA1C 0008 01 0000 60 EA 00 00 30 75 00 00 B8 0B 00 00 D0 07 00 00 F4 01
00 00 90 01 00 0004 03 02 01

Set module parameters (trigger threshold00-05)

FD FC FB FA24 0007 00 10 00 60 EA 00 00 11 00 30 75 00 00 12 00 B8 0B 00 00 13 00 D0 07 00 00 14 00 F4

01 00 00 15 00 90 0104 03 02 01 Successful reply:FD FC FB FA04 0007 01

0000 04 03 02 01

Debug output mode

Baotou	data	End of package
0xAA, 0xBF, 0x10, 0x14	20 * 16 * 4 bytes (20Doppler,16individualrange, each point 4Byte, data value of each point4byte)	0xFD, 0xFC, 0xFB, 0xFA

Energy value output mode

Baotou	data				End of package
	Data length	result	Target distance	Each distance gate energy	
0xF4, 0xF3, 0xF2, 0xF1	0x0023	No one (1byte)	2 bytes	16 *2 bytes	0xF8, 0xF7, 0xF6, 0xF5